



Session 04: The Mole & Molar Mass

Topic: The bridge between the visible world (grams) and the invisible world (atoms).

Time: 150 minutes

Big Fat Notebook: Chapter 6

Opening Attack

Try this:

Which contains more atoms: **10.0 g of helium (He)** or **10.0 g of calcium (Ca)**?

Don't guess. Prove it with numbers.

A single grain of table salt ($NaCl$) has a mass of about 2.7×10^{-5} g. Approximately how many formula units of $NaCl$ are in that one grain?

Stuck? Good. By the end of this session, you'll convert between grams, moles, and number of particles faster than you can say "Avogadro."

Worked Pattern

Example 1 — Molar Mass from the Periodic Table

Problem: What is the molar mass of helium (He)? Of water (H_2O , "water")? Of calcium carbonate ($CaCO_3$, "calcium carbonate")?

Helium (He):

From the periodic table: $\text{He} = 4.003 \text{ g/mol}$.

Molar mass = 4.003 g/mol .

→ One mole of helium atoms has a mass of 4.003 g .

Water (H_2O):

$$\text{H: } 2 \times 1.008 = 2.016 \text{ g/mol}$$

$$\text{O: } 1 \times 16.00 = 16.00 \text{ g/mol}$$

$$\text{Total: } 2.016 + 16.00 = \mathbf{18.016 \text{ g/mol}}$$

→ One mole of water molecules has a mass of 18.016 g .

Calcium carbonate (CaCO_3):

$$\text{Ca: } 1 \times 40.08 = 40.08 \text{ g/mol}$$

$$\text{C: } 1 \times 12.01 = 12.01 \text{ g/mol}$$

$$\text{O: } 3 \times 16.00 = 48.00 \text{ g/mol}$$

$$\text{Total: } 40.08 + 12.01 + 48.00 = \mathbf{100.09 \text{ g/mol}}$$

→ One mole of CaCO_3 formula units has a mass of 100.09 g .

Key insight: Molar mass is the mass of 6.022×10^{23} particles of that substance.

Molar mass has units of g/mol — always.

Example 2 — Grams → Moles (One Step)

Problem: How many moles are in 50.0 g of H_2O ?

Step 1: Find the molar mass.

$$H_2O = 2(1.008) + 16.00 = 18.016 \text{ g/mol}$$

Step 2: Divide grams by molar mass.

$$\text{mol} = 50.0 \text{ g} \div 18.016 \text{ g/mol}$$

Step 3: Calculate and apply sig figs.

$$50.0 \div 18.016 = 2.77531\dots$$

Sig figs: 50.0 (3 sig figs) limits the answer.

$$\text{mol} = 2.78 \text{ mol } H_2O$$

Answer: 2.78 mol H_2O .

$$\underbrace{50.0 \text{ g}}_{\text{grams}} \xrightarrow{\div 18.016 \text{ g/mol}} \underbrace{2.78 \text{ mol}}_{\text{moles}} \xrightarrow{\times N_A} \underbrace{1.67 \times 10^{24}}_{\text{molecules}}$$

This chain — grams → moles → particles — is the backbone of all stoichiometry.

Example 3 — Moles → Grams (Reverse Direction)

Problem: What is the mass of 0.250 mol of CO_2 (carbon dioxide)?

Step 1: Find the molar mass.

$$CO_2 = 12.01 + 2(16.00) = 44.01 \text{ g/mol}$$

Step 2: Multiply moles by molar mass.

$$\text{grams} = 0.250 \text{ mol} \times 44.01 \text{ g/mol}$$

Step 3: Calculate.

$$0.250 \times 44.01 = 11.0025$$

Sig figs: 0.250 (3 sig figs) limits the answer.

$$\text{mass} = 11.0 \text{ g } CO_2$$

Answer: 11.0 g CO_2 .

The direction signpost:

$$\begin{array}{ccccc} \text{grams} & \xrightarrow{\div \text{ molar mass}} & \text{moles} & \xrightarrow{\times \text{ molar mass}} & \text{grams} \\ g \div \frac{g}{\text{mol}} = \text{mol} & & \text{mol} \times \frac{g}{\text{mol}} = g & & \end{array}$$

The unit cancellation tells you which operation is correct.

Example 4 — Moles → Number of Particles

Problem: How many atoms are in 2.00 mol of helium?

Step 1: Recall Avogadro's number.

$$N_A = 6.022 \times 10^{23} \text{ particles/mol}$$

"1 mole of anything contains 6.022×10^{23} of those things."

Step 2: Multiply moles by Avogadro's number.

$$\text{atoms} = 2.00 \text{ mol} \times (6.022 \times 10^{23} \text{ atoms/mol})$$

Step 3: Calculate.

$$2.00 \times 6.022 \times 10^{23} = 1.2044 \times 10^{24}$$

Sig figs: 2.00 (3 sig figs).

$$\text{atoms} = 1.20 \times 10^{24} \text{ He atoms}$$

Answer: $1.20 \times 10^{24} \text{ He atoms}$.

$$\text{mol} \xrightarrow{\times 6.022 \times 10^{23}} \text{ particles}$$

Example 5 — Number of Particles → Moles

Problem: A sample contains 3.01×10^{23} molecules of O_2 (oxygen gas). How many moles is this?

Step 1: Divide number of particles by Avogadro's number.

$$\text{mol} = (3.01 \times 10^{23} \text{ particles}) \div (6.022 \times 10^{23} \text{ particles/mol})$$

Step 2: Calculate.

$$3.01 \times 10^{23} \div 6.022 \times 10^{23} = \frac{3.01}{6.022} = 0.49983...$$

The 10^{23} cancels cleanly.

Step 3: Apply sig figs.

3.01 (3 sig figs).

$$\text{mol} = 0.500 \text{ mol } O_2$$

Answer: $0.500 \text{ mol } O_2$.

$\therefore 3.01 \times 10^{23}$ is exactly $\frac{1}{2}N_A$ — half a mole. The numbers confirm it.

THE TRAP — Using the Wrong Molar Mass for Diatomic Gases

The wrong way:

How many moles are in 32.0 g of oxygen gas?

Wrong work: Molar mass of O = 16.00 g/mol.

$$\text{mol} = 32.0 \div 16.00 = \textcolor{red}{2.00 \text{ mol}} \leftarrow \text{WRONG!}$$

Why is this wrong? Oxygen gas exists as O_2 molecules, not isolated O atoms. The substance in the problem is O_2 , so you must use the molar mass of O_2 : $2 \times 16.00 = 32.00 \text{ g/mol}$.

The right way:

Step 1: Identify the substance. "Oxygen gas" = O_2 , NOT O .

$$\text{Molar mass of } O_2 = 2 \times 16.00 = 32.00 \text{ g/mol.}$$

Step 2: Divide grams by molar mass.

$$\text{mol} = 32.0 \text{ g} \div 32.00 \text{ g/mol} = 1.00 \text{ mol } O_2$$

Step 3: Verify: 32.0 g of O_2 is 1.00 mol of O_2 molecules.

$$\text{If you needed mol of O atoms: } 1.00 \text{ mol } O_2 \times 2 = 2.00 \text{ mol O atoms.}$$

Spot the difference: "Oxygen gas" means O_2 (molar mass = 32.00 g/mol). "Oxygen atoms" means O (molar mass = 16.00 g/mol). The diatomic elements are: $H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2$. Always use the molecular form for these when they appear as pure substances.

Example 6 — The Full Chain: Grams → Moles → Number of Atoms

Problem (the opening attack): Which contains more atoms: 10.0 g of helium or 10.0 g of calcium?

Helium path:

$$\text{Step 1: mol He} = 10.0 \text{ g} \div 4.003 \text{ g/mol} = 2.4981 \text{ mol He}$$

$$\text{Step 2: atoms He} = 2.4981 \text{ mol} \times (6.022 \times 10^{23} \text{ atoms/mol}) = 1.504 \times 10^{24} \text{ He atoms}$$

Calcium path:

$$\text{Step 1: mol Ca} = 10.0 \text{ g} \div 40.08 \text{ g/mol} = 0.24950 \text{ mol Ca}$$

$$\text{Step 2: atoms Ca} = 0.24950 \text{ mol} \times (6.022 \times 10^{23} \text{ atoms/mol}) = 1.502 \times$$

10^{23} Ca atoms

Answer: 1.50×10^{24} He atoms — **10 times more** than 1.50×10^{23} Ca atoms.

$$\underbrace{10.0 \text{ g He}}_{\text{same mass}} \xrightarrow{\div 4.003} \underbrace{2.50 \text{ mol}}_{10 \times \text{ more moles!}} \xrightarrow{\times N_A} 1.50 \times 10^{24} \text{ atoms}$$

$$\underbrace{10.0 \text{ g Ca}}_{\text{same mass}} \xrightarrow{\div 40.08} \underbrace{0.250 \text{ mol}}_{10 \times \text{ fewer moles}} \xrightarrow{\times N_A} 1.50 \times 10^{23} \text{ atoms}$$

$$\therefore \text{He wins by a factor of } \frac{40.08}{4.003} \approx 10$$

Example 7 — Molecules vs. Atoms Within a Compound

Problem: How many oxygen atoms are in 50.0 g of CaCO_3 ?

Step 1: $\text{mol CaCO}_3 = 50.0 \text{ g} \div 100.09 \text{ g/mol} = 0.49955 \text{ mol CaCO}_3$

Step 2: Each CaCO_3 formula unit contains 3 oxygen atoms.

$$\text{mol O atoms} = 0.49955 \text{ mol CaCO}_3 \times 3 = 1.4987 \text{ mol O atoms}$$

Step 3: $\text{atoms O} = 1.4987 \text{ mol} \times (6.022 \times 10^{23} \text{ atoms/mol}) = 9.025 \times 10^{23} \text{ O atoms}$

$$\text{Sig figs: } 50.0 \text{ (3 sig figs)} \rightarrow 9.02 \times 10^{23} \text{ O atoms}$$

Answer: 9.02×10^{23} oxygen atoms.

$$\text{grams CaCO}_3 \xrightarrow{\div 100.09} \text{mol CaCO}_3 \xrightarrow{\times 3} \text{mol O atoms} \xrightarrow{\times N_A} \text{O atoms}$$

The subscript is a mole ratio too: 1 mol CaCO_3 : 3 mol O.

Example 8 — Mass of a Single Atom/Molecule

Problem: What is the mass (in grams) of a single water molecule?

Step 1: Molar mass of $H_2O = 18.016 \text{ g/mol}$.

This is the mass of 6.022×10^{23} water molecules.

Step 2: Divide by Avogadro's number.

$$\begin{aligned}\text{mass of 1 molecule} &= 18.016 \text{ g/mol} \div 6.022 \times 10^{23} \text{ molecules/mol} \\ &= 2.9917 \times 10^{-23} \text{ g}\end{aligned}$$

Step 3: Apply sig figs.

18.016 (5 sig figs) limits the answer.

$$\text{mass} = 2.9917 \times 10^{-23} \text{ g per molecule}$$

Answer: $2.992 \times 10^{-23} \text{ g per } H_2O \text{ molecule}$.

$$\frac{18.016 \text{ g}}{\text{mol}} \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ molecules}} = 2.992 \times 10^{-23} \text{ g/molecule}$$

That is 0.0000000000000000000000002992 g — unimaginably tiny. Avogadro's number is the bridge from this microscopic world to the gram quantities we can weigh on a balance.

What We Just Did — The Mole Conversion Procedure

Step 1: Identify your starting point and your destination.

Starting: grams? particles? moles?

Destination: grams? particles? moles?

This tells you which direction to go.

Step 2: Use the correct conversion factor.

grams $\leftrightarrow$ **moles**: use molar mass (g/mol) from the periodic table.

moles $\leftrightarrow$ **particles**: use Avogadro's number (6.022×10^{23} particles/mol).

Going grams $\rightarrow$ **moles** $\rightarrow$ **particles**: chain both conversions.

Step 3: Let the units guide you.

$$\text{g} \div (\text{g/mol}) = \text{mol} \checkmark$$

$$\text{mol} \times (\text{g/mol}) = \text{g} \checkmark$$

$$\text{mol} \times (\text{particles/mol}) = \text{particles} \checkmark$$

If the units don't cancel to your destination, you used the wrong operation.

$$\text{grams} \xrightarrow{\div M} \text{moles} \xrightarrow{\times N_A} \text{particles}$$

$$\text{particles} \xrightarrow{\div N_A} \text{moles} \xrightarrow{\times M} \text{grams}$$

where M = molar mass (g/mol) and $N_A = 6.022 \times 10^{23}$ particles/mol.

Copy this. The mole is the central hub — every road from grams to particles passes through it.

Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — Grams $\leftrightarrow$ Moles direction. If the number you're converting is **bigger than the molar mass**, you have **more than 1 mole**. If smaller, less than 1 mole. This sanity check catches most calculator errors.

$50.0 \text{ g } H_2O \div 18.016 \text{ g/mol} \rightarrow$ must be $< 3 \text{ mol}$ (since $3 \times 18 = 54 > 50$). Actual: 2.78 mol. $\checkmark$

Tip 2 — Diatomic mnemonic. "Have No Fear Of Ice Cold Brew" →

$H_2, N_2, F_2, O_2, I_2, Cl_2, Br_2$. If the problem says "oxygen gas" or "nitrogen gas" and you forgot to double the molar mass, your moles will be off by a factor of 2.

Tip 3 — Avogadro shortcuts. Half a mole = 3.011×10^{23} . A tenth of a mole = 6.022×10^{22} .

These round numbers appear on multiple-choice questions constantly.

Tip 4 — Molar mass of common compounds. Memorize these — they appear in nearly every problem: $H_2O = 18.02$, $CO_2 = 44.01$, $NaCl = 58.44$, $CaCO_3 = 100.1$, $NH_3 = 17.03$, $O_2 = 32.00$, $H_2 = 2.02$ g/mol.

Tip 5 — Calculator hygiene. Keep all 10 digits on your calculator screen until the very last step. Rounding intermediate values (e.g., rounding mol from 2.7758 to 2.78 before multiplying by N_A) can shift your final answer by 10^{21} particles. Only round the final answer.

Basic Drills (B1–B10)

Goal: Automate the grams ↔ moles ↔ particles pipeline.

B1. How many moles are in 25.0 g of $NaCl$ (sodium chloride / table salt)? (Na = 22.99, Cl = 35.45 g/mol)

B2. What is the mass of 0.750 mol of $C_6H_{12}O_6$ (glucose)? (C = 12.01, H = 1.008, O = 16.00 g/mol)

B3. How many atoms are in 3.50 mol of aluminum (Al)?

B4. A sample contains 4.52×10^{22} molecules of CO_2 . How many moles is this?

B5. How many moles are in 100.0 g of $CaCO_3$? (Ca = 40.08, C = 12.01, O = 16.00 g/mol)

B6. What is the mass of 1.20×10^{24} atoms of iron (Fe)? (Fe = 55.85 g/mol)

B7. (Word problem) A gold ring contains 5.00 g of pure gold (Au = 197.0 g/mol). How many gold atoms are in the ring?

B8. (Diatomic pitfall check) How many moles are in 48.0 g of oxygen gas (O_2 , "oh-two")?

B9. (Two-part) Part 1: How many moles are in 10.0 g of H_2SO_4 ? (H = 1.008, S = 32.07, O = 16.00 g/mol) Part 2: Using that result, how many oxygen atoms are in the sample?

B10. (Speed round — under 2 min) Which contains more molecules: 5.00 g of H_2 (hydrogen gas) or 5.00 g of N_2 (nitrogen gas)? Show both calculations.

Advanced Drills (A1–A10)

Goal: Multi-step integration with Sessions 01–03. Exam traps.

A1. (Multi-step with Session 03) A compound is 40.0% C, 6.71% H, and 53.3% O by mass. (a) Find the empirical formula. (b) How many moles of this compound are in a 90.0 g sample? (Use the empirical formula mass for now.)

A2. (Multi-step with Session 02) The density of ethanol (C_2H_5OH) is 0.789 g/mL. (a) Calculate the mass of 50.0 mL of ethanol. (b) How many moles of ethanol is this? (c) How many ethanol molecules?

A3. (Trap-laden) A student says: "The molar mass of nitrogen gas is 14.01 g/mol, so 28.02 g of N_2 is 2.00 mol." What's wrong with this statement? Correct it.

A4. (Reverse engineering) A student calculates that 10.0 g of H_2O contains 3.34×10^{23} molecules. Their lab partner, rechecking, gets 5.56×10^{23} molecules. One is wrong. (a) Which one? (b) What mistake did the wrong student make? (c) Show the correct calculation.

A5. (Constructive) Write an exam problem that requires converting grams $\rightarrow$ moles $\rightarrow$ number of atoms, where the substance is a diatomic gas. Solve it. Include a note on why using the monatomic molar mass would give the wrong answer.

A6. (Constructive) Write a problem where the answer is exactly 3.011×10^{23} molecules. (Hint: this is exactly 0.500 mol.) Solve it and show why this number is significant.

A7. (Data table) A student measures five samples:

Sample	Substance	Mass (g)
A	He (helium)	2.00
B	O_2 (oxygen)	32.00
C	H_2O (water)	18.02
D	$C_{12}H_{22}O_{11}$ (sucrose)	342.3
E	Fe (iron)	55.85

For each sample: (a) Calculate moles. (b) Which two samples contain the same number of moles? Why? (c) Which sample contains the most atoms total?

A8. (Timed — under 3 min) A vitamin C tablet contains 500.0 mg of ascorbic acid ($C_6H_8O_6$). (a) Convert 500.0 mg to grams. (b) Calculate moles of $C_6H_8O_6$. (c) How many molecules of $C_6H_8O_6$ are in one tablet? (d) How many carbon atoms are in one tablet?

A9. (Hardest variation) A 2.500 g sample of a pure compound with formula MX_2 (M is a metal, X is chlorine / Cl) contains 0.01562 mol of the compound. (a) Calculate the molar mass of MX_2 . (b) Given that the molar mass of Cl is 35.45 g/mol, calculate the molar mass of M. (c) Identify the metal M from this list: Mg (24.31), Ca (40.08), Ba (137.3), Pb (207.2 g/mol). (d) Write the full formula of the compound.

A10. (Exam boss) A 5.000 g sample of hemoglobin (the oxygen-carrying protein in blood) contains 0.00340 g of iron. The molar mass of hemoglobin is approximately 6.5×10^4 g/mol. (a) How many moles of iron are in the sample? (b) How many moles of hemoglobin are in the sample? (c) How many iron atoms are in one hemoglobin molecule? (d) Hemoglobin's iron content is exactly 4 atoms per molecule. Based on this, calculate a more precise molar mass for hemoglobin. Use correct sig figs throughout.

Now Read

You can now convert between grams, moles, and number of particles in your sleep. 10.0 g of helium vs. calcium? You proved helium wins — by a factor of 10. The mole is no longer an abstraction; it's the road you travel from the balance to the atom.

Read **Chapter 6** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

You've solved 20 problems, from "how many moles in 25.0 g of NaCl" to "how many iron atoms in one hemoglobin molecule." The concepts — what a mole actually means, why Avogadro's number exists, how molar mass connects the periodic table to the lab bench — will lock into place.

Solutions: [solutions/04-solutions.md](https://www.khanacademy.org/a/solutions/04-solutions.md)



Key Terms — Quick Reference

Term	What it means
mole (mol)	6.022×10^{23} things — the chemist's "dozen"
Avogadro's number (N_A)	6.022×10^{23} particles/mol
molar mass (M)	mass of 1 mole in g/mol; found on the periodic table
grams → moles	divide by molar mass: $\text{mol} = \text{g} \div M$
moles → grams	multiply by molar mass: $\text{g} = \text{mol} \times M$
moles → particles	multiply by N_A
particles → moles	divide by N_A
diatomic elements	$H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2$ — always use molecular molar mass
formula unit	the smallest repeating unit of an ionic compound (e.g., NaCl, not a molecule)
molar mass of hydrate	includes the mass of water molecules ($CuSO_4 \cdot 5H_2O = 249.7$ g/mol)